

RAHUL KUMAR JHA

AI & Generative AI Engineer | Agentic Systems & LLM Applications

Delhi, India • +91-8383933767 • rahulkumarjha26@gmail.com • rahuljha.co.in • github.com/rahulkumarjha26 • linkedin.com/in/rahulkumarjha26

PROFESSIONAL SUMMARY

AI & Generative AI Engineer with 5 years of experience architecting enterprise-grade LLM applications, Agentic RAG workflows, automated computer vision pipelines, and resilient backend microservices. Proven track record in building multi-agent systems, optimizing inference workloads, and deploying scalable services across Azure and AWS for global enterprise clients. Skilled in prompt optimization, context management, semantic retrieval, and low-latency API serving.

TECHNICAL SKILLS

Languages & Databases: Python, SQL, DuckDB, MySQL, PostgreSQL, Redis

Applied ML & Modeling: Statistical Modeling, Predictive Analytics, EDA, Scikit-learn, Pandas, NumPy, Feature Engineering, XGBoost, Anomaly Detection

Deep Learning & CV: PyTorch, TensorFlow, Keras, OpenCV, OCR (TrOCR/Tesseract), Image & Video Processing, Transfer Learning, CNNs

Generative AI & LLMs: Agentic AI, LangGraph, LangChain, RAG Pipelines, Local LLM Inference (Ollama), Vector DBs (Pinecone, Chroma), Prompt Engineering, MLflow

Cloud & Deployment: Azure (AI Foundry, Functions, Blob, AKS), AWS (S3, CloudWatch, ECS, Lambda), Docker, FastAPI, REST APIs, Git

PROFESSIONAL EXPERIENCE

Senior Developer – Agentic Systems & RAG | WPP Production (formerly Hogarth Studios)

Aug 2024 – Jul 2026 | Gurugram, India

- Architected and deployed an enterprise **Agentic RAG system** leveraging Azure AI Foundry, OpenAI, LangChain, and vector databases for brand compliance and marketing intelligence, cutting search latency by **45%**.
- Engineered multi-modal Generative AI workflows pairing computer vision (OpenCV) with LLM-driven structured extraction to automate creative asset tagging and regulatory validation across 10,000+ daily assets.
- Built low-latency **FastAPI and Docker** microservices for production LLM serving, implementing prompt caching and context window optimization to minimize inference latency and compute costs.
- Implemented **MLflow** for LLM tracing, prompt versioning, and automated evaluation metrics (faithfulness, answer relevance), reducing container execution overhead by **30%**.

Generative AI Engineer – LLM Applications & NLP | Cognizant

Feb 2023 – May 2024 | Bengaluru, India

- Engineered internal **Generative AI and NLP solutions** for global banking clients, automating extraction, categorization, and summarization of complex financial transaction logs and compliance policies.
- Developed semantic search and retrieval pipelines utilizing vector embeddings and similarity indexing to surface suspicious transaction narratives and fraud indicators.
- Collaborated with risk and compliance teams to operationalize automated fraud-scoring logic, verifying LLM outputs against ground-truth account records to improve anomaly detection precision.

Associate AI Developer – NLP & Intelligent Automation | Cognizant

Jan 2022 – Jan 2023 | Bengaluru, India

- Engineered automated **NLP text extraction and data ingestion pipelines** utilizing Python, AWS (S3, CloudWatch, ECS), and MySQL to process complex banking transaction records, cutting report generation latency by **55%**.
- Constructed intelligent auditing services and parsing engines, replacing manual auditing workflows with automated text and anomaly validation for risk compliance teams.
- Designed modular validation modules and automated testing routines, accelerating client AI automation rollout timelines by **25%**.

Software Engineer – Data & Platform | Turbo Hire

Jul 2021 – Jan 2022 | Hyderabad, India

- Developed data ingestion and processing workflows for an AI-powered talent intelligence platform, handling thousands of unstructured candidate profiles daily.
- Integrated RESTful APIs and machine learning inference endpoints for automated candidate screening, entity extraction, and skill relevance scoring.
- Optimized complex SQL queries and relational database schemas in PostgreSQL, achieving a **35% improvement** in candidate search query latency.
- Implemented data validation, deduplication, and cleaning scripts to ensure high data integrity across relational talent pools.

KEY PROJECTS

Numera — Air-Gapped Tabular Intelligence Engine | github.com/rahulkumarjha26/numera

Open Source / Featured

- Python 3.12, DuckDB OLAP, LangGraph, Ollama (Local LLM), Next.js 16, Polars, FastAPI
- Architected an air-gapped enterprise tabular intelligence engine allowing confidential analysis of 50,000+ row spreadsheets with zero cloud data transmission and zero context overflow.
 - Implemented schema virtualization and deterministic in-memory DuckDB OLAP execution, reducing token overhead by **99.8%** and completing complex analytical aggregations in **under 20ms**.

Threadmark — AI Insurance Command Center | github.com/rahulkumarjha26/threadmark

Open Source / Featured

- Python 3.12, FastAPI, Postgres + pgvector, Next.js 15, Hybrid RAG (RRF), Cross-Encoder, Evals
- Architected a multi-agent system (6 specialist agents) where financial computations and eligibility are deterministic code tools and citations are verified in code before display.
 - Implemented a 5-layer retrieval pipeline (dense vectors, BM25 hybrid with Reciprocal Rank Fusion, cross-encoder rerank) achieving **94.2% refusal accuracy** across an 86-case golden evaluation benchmark.

RetailGuard AI — Loss Prevention & Theft Vision Engine | github.com/rahulkumarjha26/retail-guard-ai

Computer Vision / Open Source

- Python, PyTorch, YOLOv11-Pose, ViViT Transformers, OpenCV, FastAPI, Edge MPS/CUDA
- Engineered a retail theft surveillance system combining YOLOv11 human pose estimation, anatomical concealment geometry, and spatiotemporal Video Vision Transformers (ViViT).
 - Developed dynamic pocket projection anchored to 17 skeletal keypoints and an occlusion-verification state machine; sustained **30.0+ FPS** real-time inference on Apple Silicon MPS/CUDA.

EDUCATION

Bachelor of Technology in Computer Science & Engineering Chandigarh Engineering College, Landran

2017 – 2021

CERTIFICATIONS

Machine Learning Specialization — Stanford Online / Coursera
Generative AI with Large Language Models — DeepLearning.AI
AWS Certified Cloud Practitioner — Amazon Web Services
AI: Constraint Satisfaction — IIT Madras